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Melmaruvathur- 603 319, Chengalpattu (D.T), Tamilnadu, India.

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DEPARTMENT OF INFORMATION TECHNOLOGY

A-DACS: AI-POWERED DIGITAL CLEARANCE SYSTEM

Transforming Institutional Clearance through Cloud Automation and Machine Learning

PRESENTED BY

Arisha K (420422205002)

Jayasri P (420422205011)

Priya M (420422205023)

MENTOR

Mrs S. LAVANYA, M.E.

Assistant Professor/IT

ABSTRACT

Digital transformation in Academic Administration requires secure and automated **Fee Verification Systems**. Traditional **Paper-Based Clearance Processes** are prone to **Receipt Tampering**, **Manual Errors**, and **Certificate Forgery**. This research addresses critical issues such as **Fake Screenshot Manipulation**, Delayed Approvals, lack of Real-Time Parent Communication, and absence of Secure Validation Mechanisms. The proposed A-DACS System integrates **AI-Based OCR**, **Regex Validation**, and **Probability Scoring** for accurate receipt verification. A novel **Text Tampering Detection Framework** compares Original OCR Data with User-Edited Inputs to prevent fraud. Secure **QR-Based Live Certificate Verification** ensures authenticity. Compared to existing systems, the solution improves Financial Transparency, reduces Administrative Workload, and enhances Security through **Cloud Automation** and Role-Based Access Control.

INTRODUCTION

Background and Context

The rapid growth of digital technology in educational institutions has increased the need for secure and automated financial systems. However, many colleges still depend on **manual fee verification** and paper-based clearance processes. The rise of digital payments such as **UPI transactions** has also increased the risk of **receipt tampering** and data manipulation. Existing systems lack proper **fraud detection mechanisms** and secure certificate validation, creating administrative and security challenges.

Problem Statement

Most institutions continue to verify payment receipts manually, leading to errors, delays, and potential **financial fraud**. There is no structured method to detect **edited payment screenshots** or manipulated data. Additionally, traditional No-Due certificates can be forged due to the absence of **real-time verification systems**. Therefore, there is a need for an **AI-based automated clearance system** that ensures secure verification, tamper detection, and improved transparency.

LITERATURE SURVEY

Literature Review (2024–2025)

Title & Description	Technical Methodology	Advantages	Disadvantages
LMV-RPA (Abdellatif et al., 2025) Voting-based RPA for document processing.	Uses multiple OCR engines with voting and LLM interpretation.	Improves robustness and reliability in document extraction.	High resource usage; not suitable for mobile; no authenticity validation.
ERPA (Abdellatif et al., 2024) OCR integrated with LLM for automation.	Combines OCR with LLM for contextual correction.	Enhances semantic understanding and reduces manual effort.	General-purpose; lacks domain-specific validation and fraud detection.
MonkeyOCR v1.5 (Zhang et al., 2025) Vision-Language layout parsing framework.	Two-stage vision-language model for layout and text analysis.	Strong layout understanding for complex documents.	Computationally heavy; excessive for simple receipts.

Literature Review (2024–2025)

Title & Description	Technical Methodology	Advantages	Disadvantages
Multi-Stage Field Extraction (Jin et al., 2025) Financial field extraction model.	Segment-based processing using vision-language models.	Accurate key-value extraction; efficient field targeting.	Complex setup; no confidence scoring or verification layer.
Transformer-Based OCR (Khallouli et al., 2025) Augmented transformer OCR model.	Transformer OCR trained with synthetic data augmentation.	Handles diverse fonts and noisy inputs effectively.	Requires large training data; high computational demand.
Comparative Analysis (Patel, 2025) OCR and AI tool evaluation study.	Benchmarks various OCR and AI tools.	Identifies cost-effective OCR solutions.	No new system proposed; only comparative study.

EXISTING SOLUTIONS & LIMITATIONS

Existing Approach	Core Technology	Primary Limitation
Traditional OCR Portals	Single-engine (Tesseract)	Struggles with varied fonts, noisy backgrounds, and "O" vs "0" errors
Enterprise RPA	Logic-based Automation	Optimized for large corporate datasets; not tailored for education
Manual Verification	Human Audit	Slow, error-prone, no real-time tracking
Generic VLM Models	Vision-Language Models	High computational cost; overkill for simple, single-page receipts
Basic Field Extractors	Multi-stage segmentation	Requires complex pre-processing for each new format

OBJECTIVES

1. To develop a **mobile-first fee management system** using Flutter
2. Firebase serverless architecture, including **web support**.
3. To implement **OCR-based receipt verification** for accurate and secure payment validation.
4. To enable **real-time cloud synchronization** across mobile and web platforms.
5. To automate **payment reminders and verification workflows** using cloud functions.
6. To design a **Text Tampering Detection Mechanism**
7. To generate **QR-Based Secure No-Due Certificates**
8. To enhance **Financial Transparency and Fraud Prevention**

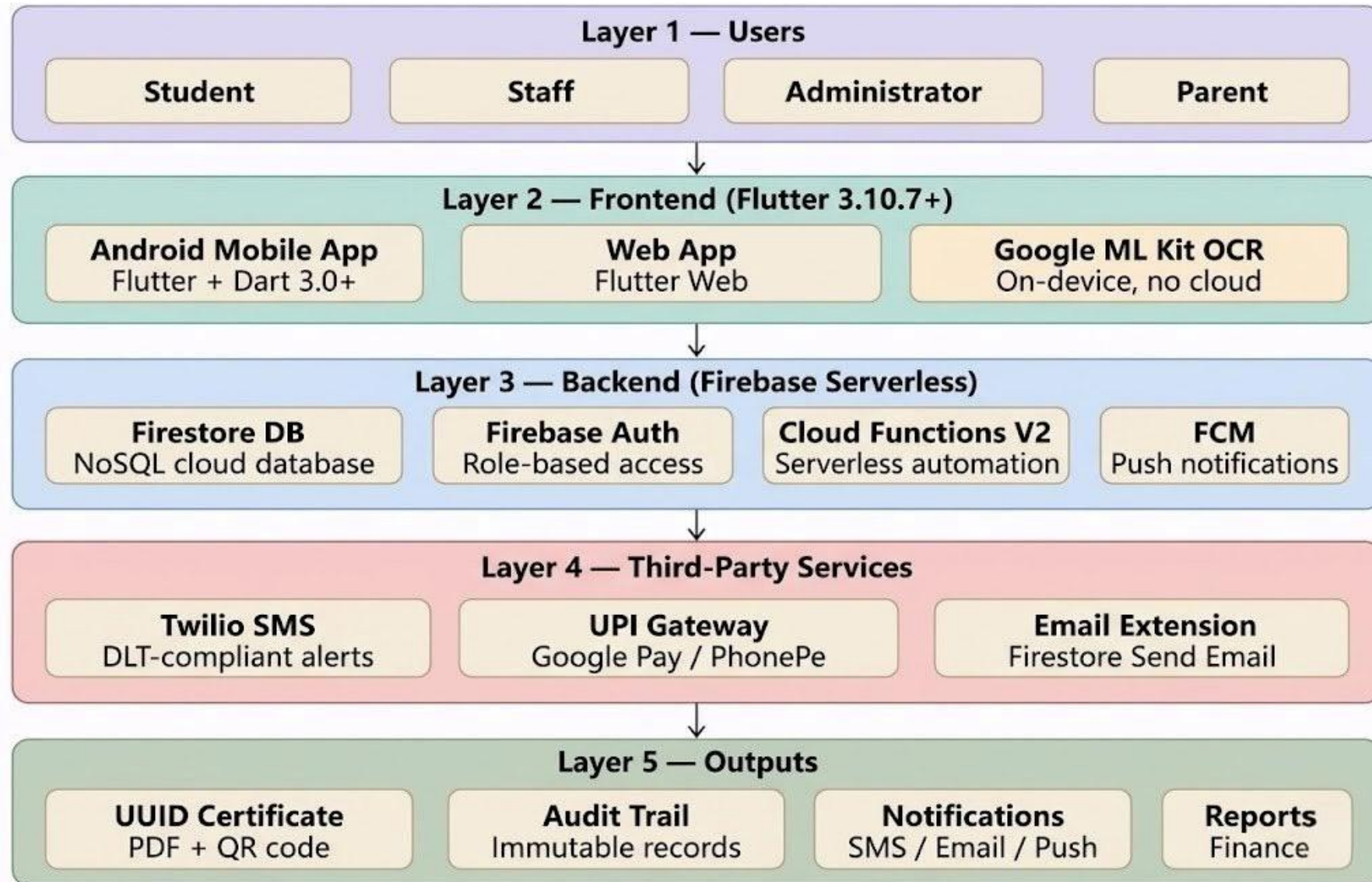
PROPOSED INNOVATION

- Development of a secure **AI-Based Digital Academic Clearance System**
- Implementation of **OCR Technology** for automatic receipt data extraction
- Integration of **Regex Validation** for structured transaction verification
- Deployment of **Cloud-Based Automation** for reminders and status updates
- Role-based access control for **Students, Staff, and Admin**

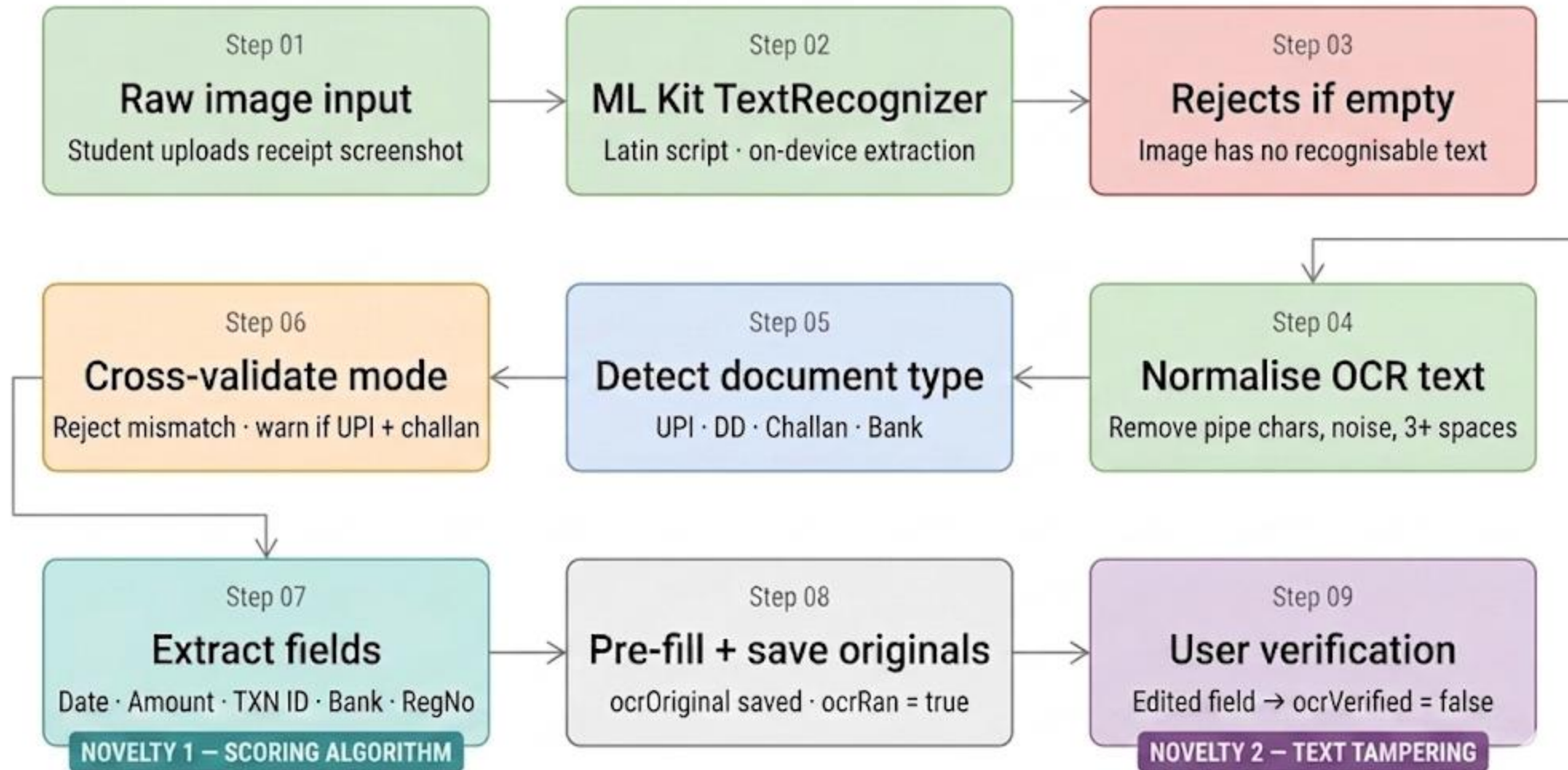
NOVELTY

- **Text Tampering Detection Mechanism (TTDM)** comparing OCR data with edited inputs
- Design of a **Probability Scoring Algorithm** for accurate amount validation
- Implementation of **Cryptographically Unique Certificate IDs (UUID)**
- Real-time **QR Verification linked to Cloud Database**
- Hybrid approach combining **AI + Context-Based Validation Logic**

SYSTEM DESIGN & ARCHITECTURE



OCR WORKFLOW (ALGORITHM)



TECHNOLOGY ARCHITECTURE

Tier	Technology & Tools		
Frontend	Flutter SDK 3.10.7+ Android mobile + Web browser Dart 3.0+ · Provider 6.1+	Google ML Kit OCR On-device text recognition No cloud — privacy-safe	
AI Engine	Probability Scoring +100 pts = exact fee match Confidence-level output	Tamper Detection OCR-original vs submitted Red flag in admin panel	Regex UPI · DD Amount
Backend	Firestore NoSQL database Audit trail	Firebase Auth Role-based Admin approval	Cloud Fn. V2 Serverless logic 24/7 automation
Comms	Twilio SMS Parent alerts EN / Tamil	FCM Push Real-time alerts Mobile + web	Email Firestore extension Automated
Payments	UPI Intent Google Pay · PhonePe External redirect — secure	QR Code Scan to pay Instant	Demand Draft DD scan upload OCR verified

MODULES

Module 1

User Management & Authentication

- Student / Staff / Admin registration
- Firebase Auth — Email
- Manual admin approval workflow
- Role-based access & routing
- System bootstrap - master admin seed
- Session management & security rules
- Student & Staff master-list pre-loading

Module 2

Payment Processing & Fraud Detection

- UPI - QR Code - Demand Draft payments
- 6-strategy OCR amount extraction
- Document-type detection (UPI/DD/Challan)
- Cross-validation & mode mismatch reject
- Scoring Algorithm (+100 / +50 / +10 / -20)
- Tamper detection — OCR vs submitted diff
- Admin payment verification & audit trail

Module 3

Certificate & Communication Management

- UUID-keyed No-Due certificate generation
- QR code embedded PDF via pdf package
- Public /verify portal — live DB check
- FCM push - Twilio SMS - Email alerts
- 7/3/1-day automated deadline reminders
- Bulk overdue SMS (admin callable)
- Reissue request & approval workflow

METHODOLOGY

Research Design

This project follows a Design and Development Research Approach using a **Case Study model** to analyze institutional fee workflows and develop a cloud-based automated solution. The system was implemented using a **Serverless Architecture (Flutter + Firebase)** to ensure scalability and security.

Data Collection Methods

Data was collected through **requirement analysis discussions** with administrative staff (Comma Separated Value), review of existing fee management procedures, generation of test datasets, and **user feedback evaluation**.

Data Analysis Techniques

A **Mixed-Method approach** was applied for analysis. **Quantitative** techniques measured system accuracy, response time, and automation efficiency, while **Qualitative** techniques assessed usability, workflow clarity, and overall system reliability.

TESTING & RESULTS

Testing Strategy

- **Unit Testing** – Verified core modules (Auth, Fee Logic, Wallet, PDF)
- **Integration Testing** – Validated Flutter–Firebase connectivity
- **Performance Testing** – Checked response time & real-time updates
- **Security Testing** – Ensured RBAC & Firestore rule enforcement

Results

Unit Testing

OCR Accuracy . 95%
Tamper Detection . 98%
Speed . <2s

Integration Testing

Flutter–Firebase . ✓ Verified
UUID unique . 100%
QR verify . 100%

Performance Testing

Clearance time . ~2 min
Admin workload . ~75%
Gen. speed . <3s

Security Testing

SMS delivery . 98%
Push alerts . <30s
Uptime . 99.5%
Fraud savings . ₹10L/yr

CONCLUSION

Summary

The A-DACS platform **automates fee management** and No-Due issuance through a **cloud-native serverless architecture** powered by Flutter and Firebase. Replacing legacy systems eliminates manual delays and significantly reduces accounting errors, ensuring a fast, scalable, and accurate administrative workflow.

Key Contributions

- **Security:** Scalable digital management with secure Firebase Authentication.
- **Logic:** Implemented complex hierarchical fee aggregation for departmental accuracy.
- **Innovation:** Automated QR-verified No-Due certificates for instant validation.
- **Engagement:** Integrated digital wallets with real-time Push & SMS notifications.

Future Roadmap

Full integration with **External Payment** Gateways (e.g., Razorpay or Stripe) for 100% automated transaction reconciliation

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- [6] Himani et al. "Design and Deployment of AI-Driven Fraud Detection Models in Cloud-Based Banking Ecosystems." In *2025 International Conference on Networks and Cryptology (NETCRYPT)*, pp. 223-228. IEEE, 2025.

PROJECT DELIVERABLES

GITHUB LINK: <https://github.com/Jayasrip08/A-DACS.git>

APK LINK: <https://drive.google.com/file/d/1arpFD6qR2AXnaKrp8j1PwlryLLVaL9L/view?usp=sharing>

WEB LINK: <https://apec-no-dues.web.app/>

COMPLETE DEMO LINK:

<https://drive.google.com/file/d/1XUfmzxDmduudY4mzaLsZatpLWiT75LLY/view?usp=sharing>

THANK You

A-DACS · AI-Powered Digital Clearance System

Arisha K | Jayasri P | Priya M

Department of Information Technology · Adhiparasakthi Engineering College

Mentor: Mrs. S. Lavanya, M.E.